Reg. No [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18CE32]

# THIRD SEMESTER B. Tech EXAMINATION, DECEMBER–2021

## SURVEYING-I

**Time: 3 Hours** **Maximum Marks: 70**

**Instructions:**
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1. a. Give a brief note on any two different Instruments Required For Linear Measurement. **06**

   b. List and Explain the Various Types Of Surveying Based On Methods. **08**

2. a. List and Explain the different steps involved in Temporary Adjustments. **06**

   b. The following are the bearings of a closed traverse using a prismatic compass. Compute the included angles and the deflection angles. **08**

| Line | AB | BC | CD | DE | EF | FA |
|---|---|---|---|---|---|---|
| **Bearing** | 37° 30′ | 92° | 151° 30′ | 220° 15′ | 283° | 330° 15′ |

3. a. List and Explain any two different Types of levels which is used in Surveying. **04**

   b. The following staff readings were observed successively with a level. The instrument having been moved after third, sixth and eight readings. **2.228, 1.606, 0.988, 2.090, 2.864, 1.262, 0.602, 1.982, 1.044, 2.684** meters. Enter the above readings in page of a level book and calculate the R. L. of points if the 1st reading was taken with a staff held on a BM of 432.384m. **10**

OR

4. a. The following consecutive readings were taken with a level and 5m leveling staff on continuously sloping ground at a common interval of 20m.
**0.385, 1.030, 1.925, 2.825, 3.730, 4.685, 0.625, 2.005, 3.110, 4.485.**
The Reduced Level of the first point was 208.125m. Rule out a page of a level field book and enter the above readings. Calculate the reduced levels of the points by Rise and Fall Method and also the Gradient of the line joining the first and the last point. **14**

5. a. Explain direct method with transiting to set out the traverse using theodolite. **06**

   b. A closed traverse was conducted round an obstacle and the following observations were made. Work out the missing quantities. **08**

| Side | Length in m | Azimuth |
|---|---|---|
| AB | 500 | 98° 30′ |
| BC | 620 | 30° 20′ |
| CD | 468 | 298° 30′ |

568

| | | |
|---|---|---|
| DE | ? | $230^0$ 00′ |
| EA | ? | $150^0$ 10′ |

**OR**

**6.** **a.** Explain the Bow-ditch's correction and Transit Rule. **08**

**b.** Explain the Direct angle and Deflection angle. **06**

**7.** **a.** Explain the Direct Method of Contouring. **07**

**b.** Explain the Indirect Methods of Contouring. **07**

**OR**

**8.** The following perpendicular offsets were taken at 10 meters intervals from a survey line to an irregular boundary line.

**3.25, 5.60, 4.20, 6.65, 8.75, 6.20, 3.25, 4.20, 5.65** **14**

Calculate the areas enclosed between all the lines using Average Ordinate Rule, Trapezoidal Rule and Simpson's Rule.